

## Subset Count (scout)

Kida decided to test your knowledge again. You are given a list  $V$ , of  $N$  integers. Kida asks you to count the number of interesting subsets that satisfy the following condition: for any two numbers  $V_x$  and  $V_y$  in the subset,  $V_x$  appears in the subset the same number of times as  $V_y$ .

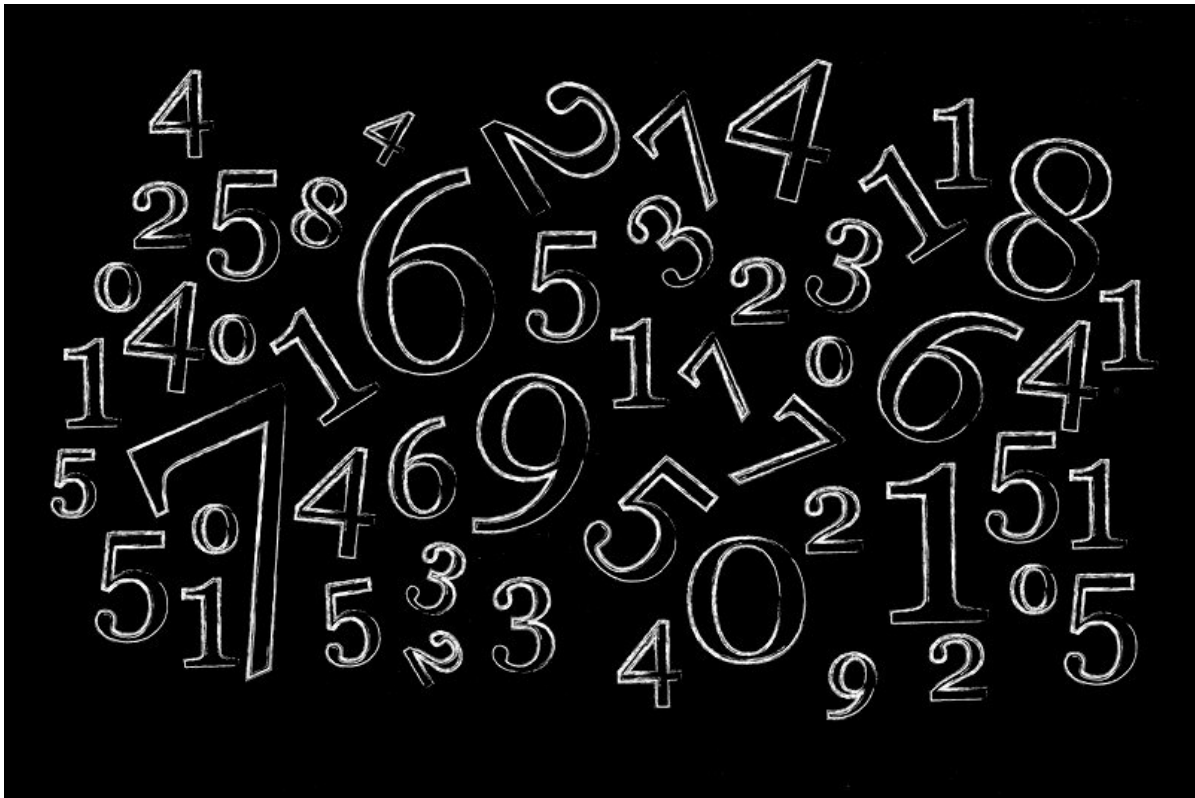


Figure 1: This is not an interesting set.

Two subsets are considered different if there is at least one value  $i$  such that  $V_i$  is in one subset but not in the other. Note that two different subsets might contain the exactly same numbers.

Since this number can be very large, you need to count it modulo  $10^9 + 7$ .

 Among the attachments of this task you may find a template file `scout.*` with a sample incomplete implementation.

### Input

The first line contains the only integer  $N$ . The second line contains  $N$  integers  $V_i$ .

### Output

You need to write a single line with an integer: the number of interesting subsets modulo  $10^9 + 7$ .

✎ The *modulo* operation ( $a \bmod m$ ) can be written in C/C++/Python as  $(a \% m)$  and in Pascal as  $(a \bmod m)$ . To avoid the *integer overflow* error, remember to reduce all partial results through the modulus, and not just the final result!  
Notice that if  $x < 10^9 + 7$ , then  $2x$  fits into a C/C++ `int` and Pascal `longint`.

## Constraints

- $1 \leq N \leq 100\,000$ .
- $1 \leq V_i \leq 1\,000\,000\,000$  for each  $i = 0 \dots N - 1$ .

## Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points)      Examples.  

- **Subtask 2** (20 points)       $1 \leq N \leq 12$ .  

- **Subtask 3** (20 points)       $1 \leq V_i \leq 2$ .  

- **Subtask 4** (30 points)       $1 \leq N \leq 1000$ .  

- **Subtask 5** (30 points)      No additional limitations.  


## Examples

| input          | output |
|----------------|--------|
| 3<br>1 2 2     | 7      |
| 5<br>1 2 2 1 3 | 21     |

## Explanation

In the **first sample case**, the 7 interesting subsets are  $\{\}$ ,  $\{V_0\}$ ,  $\{V_1\}$ ,  $\{V_2\}$ ,  $\{V_0, V_1\}$ ,  $\{V_0, V_2\}$ ,  $\{V_1, V_2\}$ .  $\{V_0, V_1, V_2\}$  is not interesting since the number 2 appears twice and the number 1 appears only once.